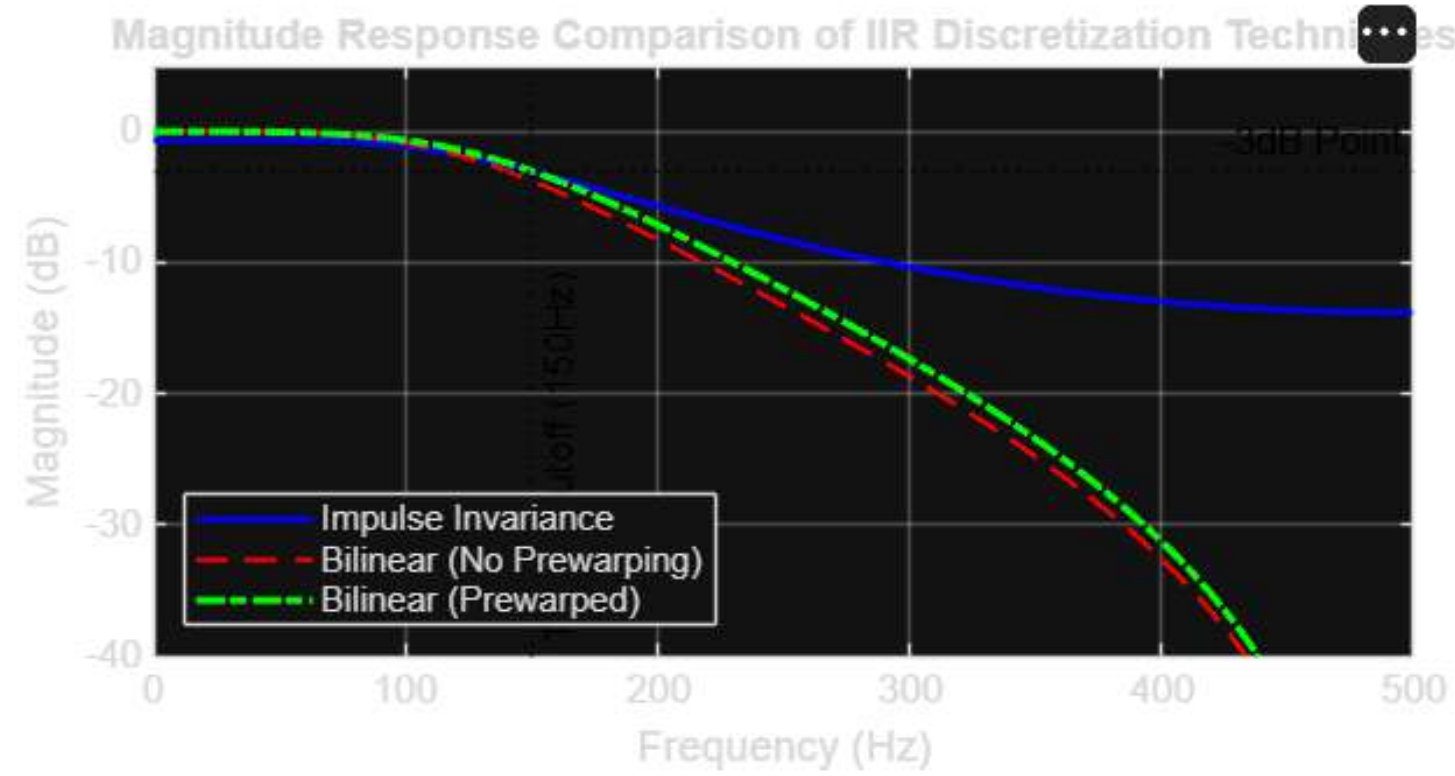
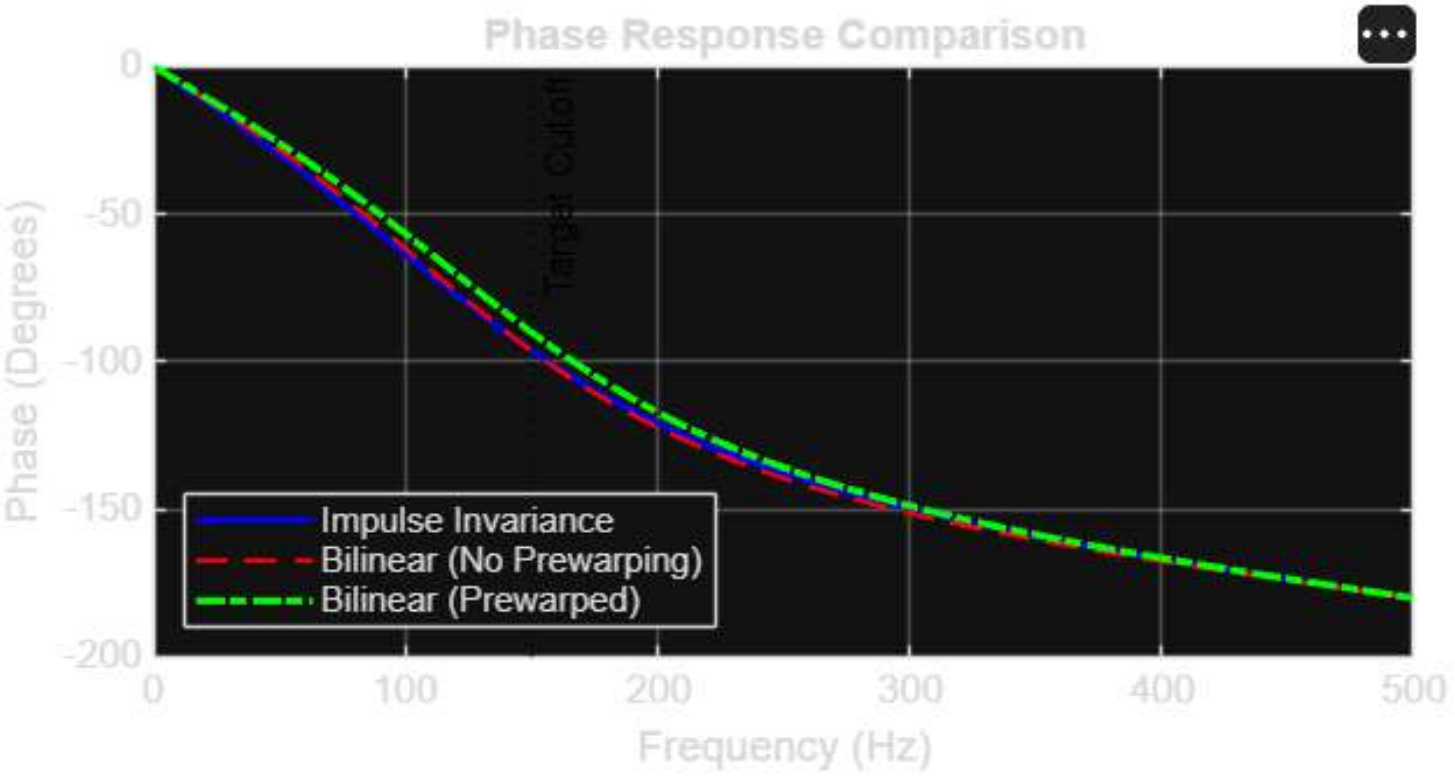
```
fs = 1000;
Ts = 1 / fs;
fc = 150;
order = 2;
w_c = 2 * pi * fc;
[B_analog, A_analog] = butter(order, w_c, 's');
Omega_prewarped = (2 / Ts) * tan((w_c * Ts) / 2);
[B_analog_prewarped, A_analog_prewarped] = butter(order, Omega_prewarped, 's');
[num_iim, den_iim] = impinvar(B_analog, A_analog, fs);
[num_blt_raw, den_blt_raw] = bilinear(B_analog, A_analog, fs);
[num_blt_pw, den_blt_pw] = bilinear(B_analog_prewarped, A_analog_prewarped, fs);
N_pts = 1024;
[H_iim, f_vec] = freqz(num_iim, den_iim, N_pts, fs);
[H_blt_raw, ~] = freqz(num_blt_raw, den_blt_raw, N_pts, fs);
[H_blt_pw, ~]  = freqz(num_blt_pw, den_blt_pw, N_pts, fs);
[~, target_idx] = min(abs(f_vec - fc));
mag_iim_db = 20*log10(abs(H_iim(target_idx)));
mag_blt_raw_db = 20*log10(abs(H_blt_raw(target_idx)));
mag_blt_pw_db = 20*log10(abs(H_blt_pw(target_idx)));
fprintf('=========================================================\n');
fprintf('         IIR FILTER DESIGN PARAMETER EVALUATION          \n');
fprintf('=========================================================\n');
fprintf('Target Cutoff Frequency: %d Hz (Target Mag: -3.01 dB)\n\n', fc);
fprintf('1. Impulse Invariance Method (IIM):\n   Magnitude at %dHz: %.2f dB\n\n', fc, mag_iim_db);
fprintf('2. Bilinear Transform (No Prewarping):\n   Magnitude at %dHz: %.2f dB\n\n', fc, mag_blt_raw_db);
fprintf('3. Bilinear Transform (With Prewarping):\n   Magnitude at %dHz: %.2f dB\n', fc, mag_blt_pw_db);
fprintf('=========================================================\n');
figure('Color', [1 1 1]);
subplot(2,1,1);
plot(f_vec, 20*log10(abs(H_iim)), 'b-', 'LineWidth', 1.5); hold on;
plot(f_vec, 20*log10(abs(H_blt_raw)), 'r--', 'LineWidth', 1.2);
```

```
plot(f_vec, 20*log10(abs(H_iim)), 'b-', 'LineWidth', 1.5); hold on;
plot(f_vec, 20*log10(abs(H_blt_raw)), 'r--', 'LineWidth', 1.2);
plot(f_vec, 20*log10(abs(H_blt_pw)), 'g-.', 'LineWidth', 1.8);
xline(fc, 'k:', 'Target Cutoff (150Hz)', 'LabelVerticalAlignment', 'bottom');
yline(-3.01, 'k:', '-3dB Point');
grid on;
title('Magnitude Response Comparison of IIR Discretization Techniques');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
legend('Impulse Invariance', 'Bilinear (No Prewarping)', 'Bilinear (Prewarped)', 'Location', 'southwest');
ylim([-40 5]);
subplot(2,1,2);
plot(f_vec, unwrap(angle(H_iim))*180/pi, 'b-', 'LineWidth', 1.5); hold on;
plot(f_vec, unwrap(angle(H_blt_raw))*180/pi, 'r--', 'LineWidth', 1.2);
plot(f_vec, unwrap(angle(H_blt_pw))*180/pi, 'g-.', 'LineWidth', 1.8);
xline(fc, 'k:', 'Target Cutoff');
grid on;
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
legend('Impulse Invariance', 'Bilinear (No Prewarping)', 'Bilinear (Prewarped)', 'Location', 'southwest');
```


Magnitude Response Comparison of IIR Discretization Techniques
-3dB Point
Target Cutoff (150Hz)
Impulse Invariance
Bilinear (No Prewarping)
Bilinear (Prewarped)
Magnitude (dB)
Frequency (Hz)
0
-10
-20
-30
-40
0
100
200
300
400
500


Phase Response Comparison
Target Cutoff
Impulse Invariance
Bilinear (No Prewarping)
Bilinear (Prewarped)
Phase (Degrees)
Frequency (Hz)
0
-50
-100
-150
-200
0
100
200
300
400
500